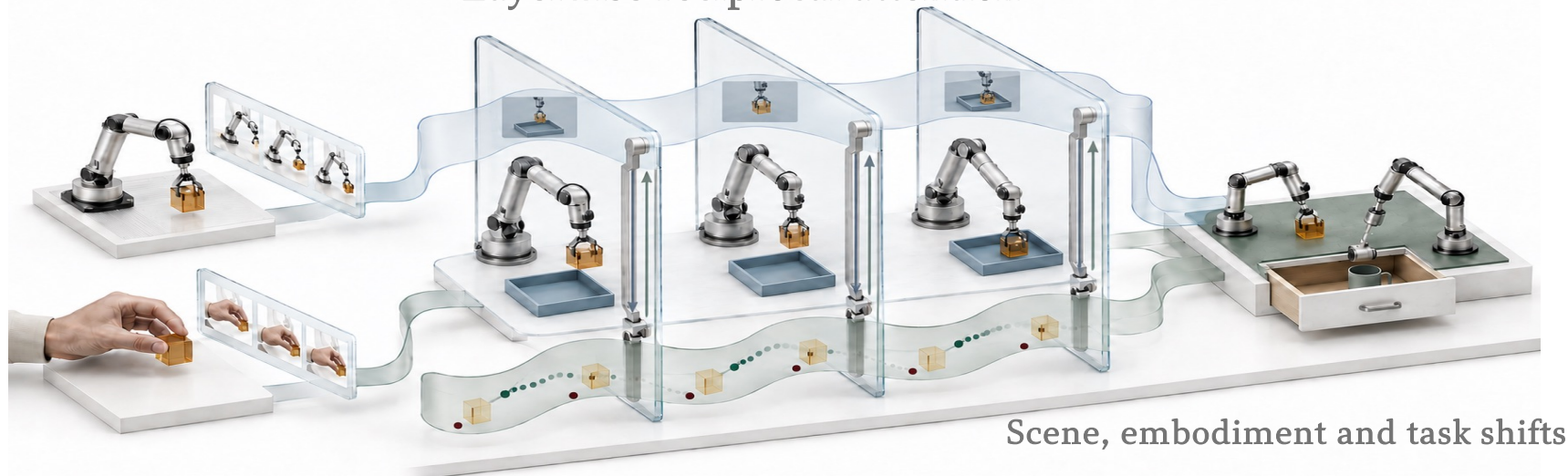


JAM: Action-Grounded World Pretraining

Layerwise reciprocal attention



Video features $\longleftrightarrow$ Action denoising
 Future-video denoising $\longleftrightarrow$ Action + effect features

Attention architecture

Query-to-key visibility

Independent One-way

JAM



Cross-stream reads

$$K = [K_w; K_a], \quad V = [V_w; V_a]$$

Separate queries, shared key/value access

Coupled video–action denoising

Video and action guide each other at every step

Noise

Joint refinement

Prediction

Video stream

Action stream

